

Toy Car

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Individual Project Report
ME 1670

Project Description:

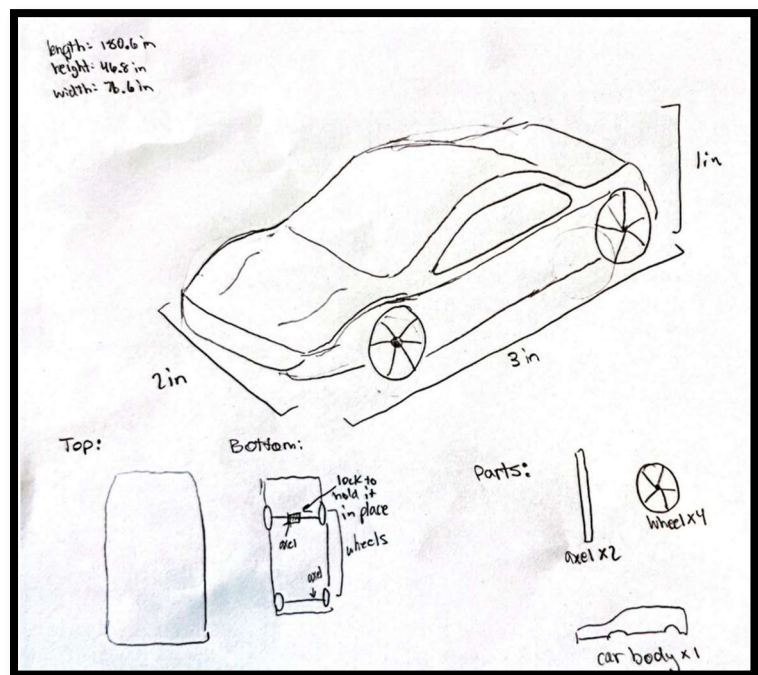
For my individual project I chose to design a toy car. I wanted something that was cool to look at that also had moving parts (the wheels).

The individual project had to fit within a 3 x 3 inch space, have 2-4 interlocking parts, and had to have 4 distinct CAD operations to create each part. Additionally, the project had to have at least one interlocking feature for each part. I chose to design a toy car which fulfilled these requirements. The original part sketch I included below has too many parts, so I modified it to have 3. I connected the wheels to the axles, and made two snap fits in the car body for the axels to fit in. The idea was for the wheels to fit into the snap fit, but also be able to move.

Concept Sketch:

The car body is made of extruded features, extruded cuts, mirrors, fillets, and a shell. The wheel and axle are made of lofted features, extrudes, extruded cuts, mirrors, pattern, and fillets.

In order to get the correct dimensions and proportions for the toy car, I researched the measurements of the average sedan and divided them to fit in the given dimensions (3 x 3 inch box). I also found a picture of a toy car that I wanted to recreate so I resized it to the correct dimensions in SolidWorks, and traced over it.



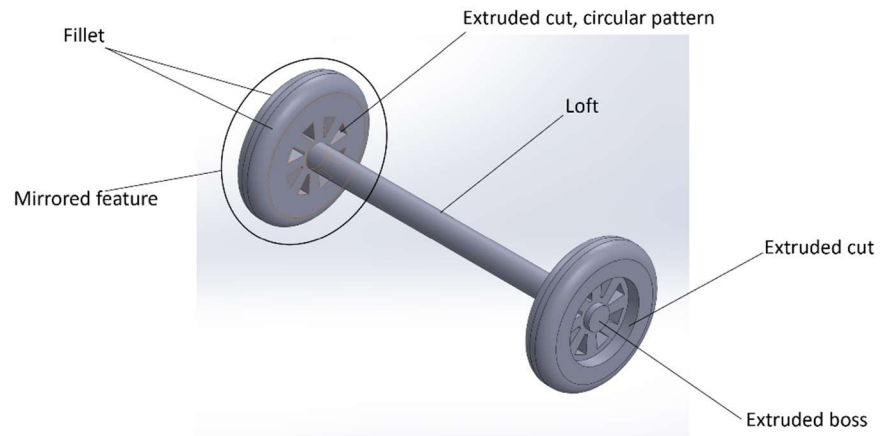
Inspiration:



CAD features:

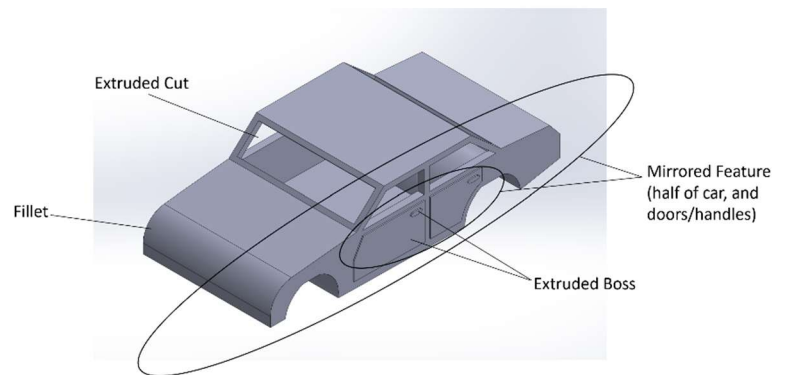
Wheel and Axle (X 2):

- Loft
- Boss-Extrude
- Cut-Extrude
- Circular Pattern
- Mirror
- Fillet



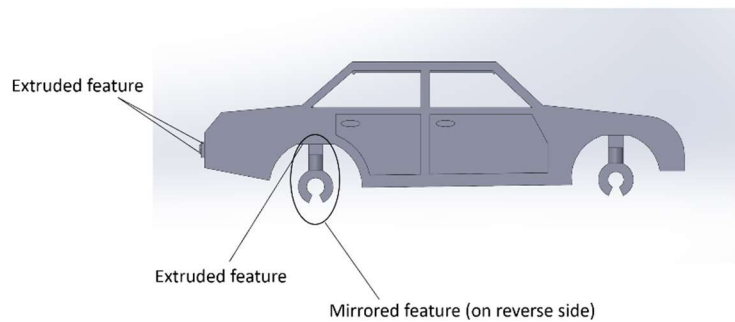
Car body:

- Boss-Extrude
- Shell
- Cut- Extrude
- Mirror
- Fillet



Interlocking features:

The hooks featured at the bottom of the car body are snap fits for the axle to fit into. There are blocks on the edges of the hollowed-out part in order to keep the wheels in place and not move side to side. This had to work for both axles for the front and back of the car, requiring 4 snap fits in total.



Design for Manufacturing Considerations:

To get the appropriate dimensions of the toy car, I researched the measurements of the average sedan's width, length, and height. Given that the maximum size that my car could be was 3 inches, I set my length to 3 inches. I then figured out the approximate dimensions of the width and height of the toy car based on the measurements of a real car.

In SolidWorks, I created two planes that were 3 inches apart to ensure I made it the appropriate size. I found an image of a toy car that I wanted to base my car off of, and I lined up the front and back of that car with my two plans. I then traced the image using lines and arcs to create the shape of my car.

I decided to use the snap fit for the wheel and axles. I chose this because I wanted the wheels to fit securely into the body of the car, but also be able to rotate. Although I did not use anything to draw inspiration from, I came up with this type of fit because I thought it was the best fit to achieve everything I wanted to do. I thought about other kinds of fits but in my opinion they did not work as well as the snap fit and would have been much more complicated to pull off. I did not include tolerances for the snap fit, because I utilized the assembly feature for putting the parts together and seeing if there was space. Tolerances is something I should have included if nothing else but to ensure that the parts would fit without difficulty and enough room to rotate.

Originally, I wanted to base my toy car after a McLaren P1 sports car, but I decided to make my design more like matchbox car design to make designing and dimensioning easier. The McLaren would have required a lot more detail to make it distinguishable. Additionally, it would have required a lot more curves and fillets, that would take a lot of time to figure out the right size, and I knew I would have to include them in the report, so I went with a simpler design.

McLaren P1:



Average Toy car:



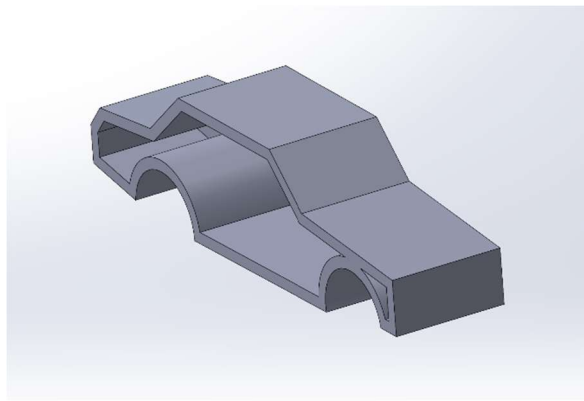
It took a lot of trial and error to get the original shape of the car right. It was tracing the picture above (right) that allowed me to be satisfied with the general shape and I could move on. I already had an idea of what I wanted to do with the snap fit at the body of the car, but it was not until I started to CAD it that I realized my restrictions and I had to try different ways to get it to work. Looking back, I had many features that were extruded, cut, and then extruded again. Doing one extruded feature with what I actually wanted to protrude would have gotten me the same

result with less complication. This was because I did not have an absolute set plan of the order in which I wanted to do things which caused a couple of unnecessary steps.

The potential problems in my design that I anticipated might happen were simply thickness issues. I knew I was making a really small and somewhat intricate design (for a 3 x 2 x 1 inch toy car) so I was not sure how much the printer could handle in terms of thickness. I made sure I was above the restriction (0.04 minimum) and I utilized the SolidWorks “evaluate” and then “Thickness” feature to see where the weakest spots in my design were so I could address them early on.

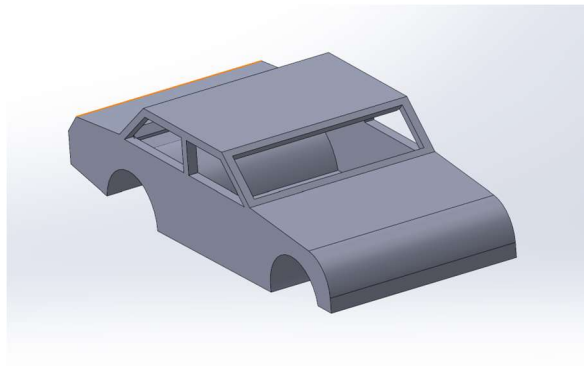
Car body:

1st Iteration:



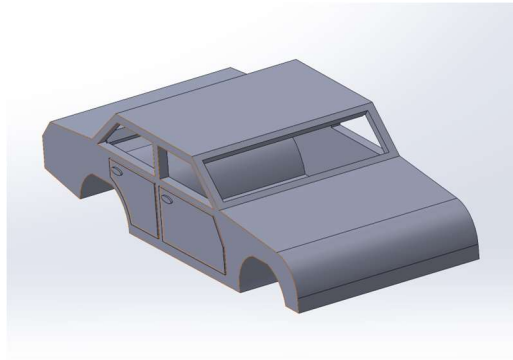
Car body is shelled.

2nd Iteration:



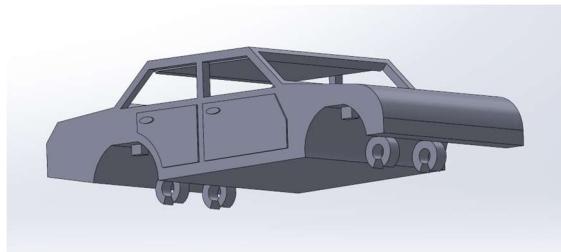
Car body mirrored along center plane.

3rd Iteration:



Door and handles added and mirrored; front of car is filleted.

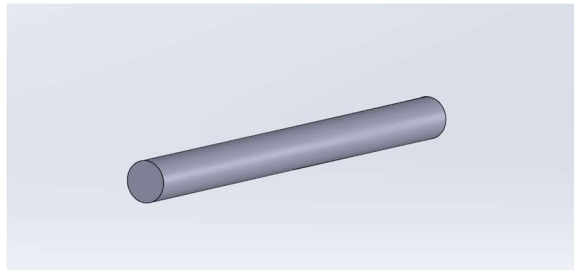
4th Iteration:



Snap fits are added at base of car, blocks to prevent wheels moving horizontally added.

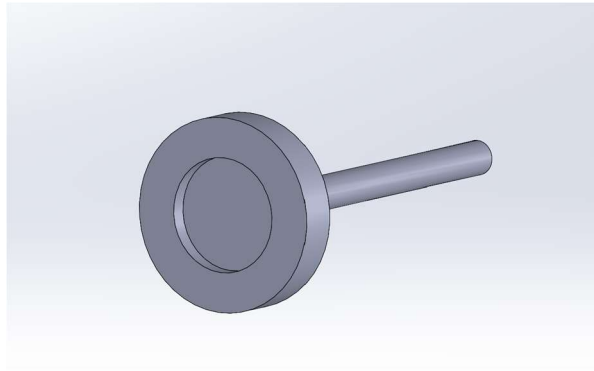
Wheel and axle:

1st Iteration:



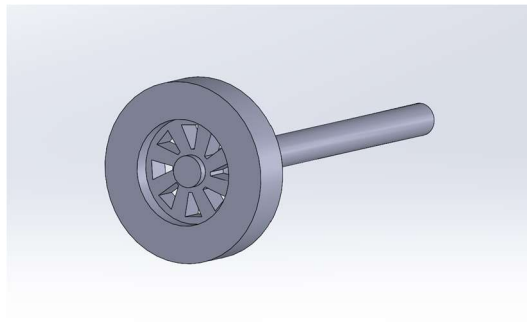
The axle was created by lofted cylinder.

2nd Iteration:



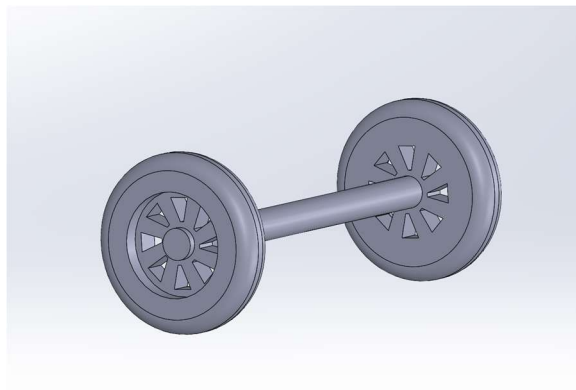
Base of wheel added.

3rd Iteration:



Extruded cut is patterned circularly around the center of the wheel.

4th Iteration:

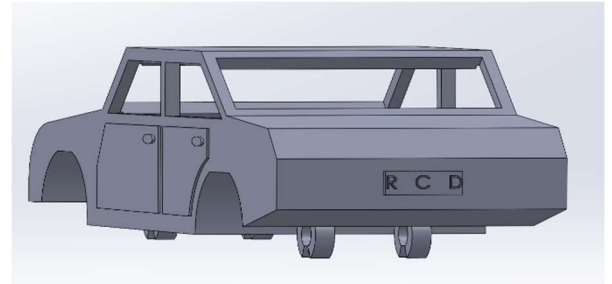


Wheel is mirrored across plane in the middle of the axle.

Challenges Faced in Manufacturing:

The car body and both wheel/axles fit after they were 3d printed. The one thing that I would change would be to add tolerances so that the axles could move more easily within the snap fit and without being forced. I will also add a little bit of Wd-40, but if it had printed with more of a gap that would have been easier.

The license plate did not show up as well as I had intended. The original design was supposed to show my initials on the plate (RCD), but when printed, it is clear there is a license plate but there are no initials. I knew this might happen simply based on the tiny dimensions of the lettering, and going forward I would probably do it a different way. I could make the letters bigger and thicker, and therefore closer together, which would be less aesthetically, but probably more practical and easier to see.



Tolerance Analysis:

Feature	Dimension in CAD (inches)	Actual Dimension (inches)	Percentage difference (%)
Wheel width	.45	.411	9.05
Length of wheel axle	1.0	1.211	19.08
Length of car	2.75	2.745	.18
Width of car	1.2	1.23	2.46
Height of car body	.71	.715	.70
Height of car + wheel	.945	.955	1.05
Height between bottom wheels to top of trunk	.695	.794	13.30
Distance between wheels (outside edge)	2.14	2.137	.14
Height of bottom of wheel to hood	.645	.619	4.11

The difference in dimensions between the original CAD design is slight. The biggest discrepancy was the length of the wheel axle, but this is likely due to the snap fit and how the axle fits within the fit. Considering the piece's parts were incredibly small and somewhat detailed, it did a great job of being 3d printed. The small details (license plate and handle) were still included and visible despite being less than 0.04 inches in thickness.

Conclusion:

Initially, the design I wanted was much more complicated and detailed. As I tried my initial sketches, I realized what was in my skillset range and what was not, and I had to limit my initial car form. One of my biggest takeaways from this project was learning what I could and

could not do and what I wanted to work on in the future. I think I could have done the design I wanted to if I had more time to learn the material, and execute it in the right way, but I ran out of time and had to stick with my simple design. The hardest part when CADing was the initial outline. Once I figured out how to use an online picture of a toy car to trace, it was easy from there on out. A couple things did give me a little trouble, and I took the long way for a lot of parts. I mirrored a couple features separately when I could have waited and done it all at once. For the snap fit, I wanted to mirror it across the halfway point of the car body, but the back snap fit needed to be longer than the front. Most of the problems I encountered were not complicated, but just required time.

Drawings:

